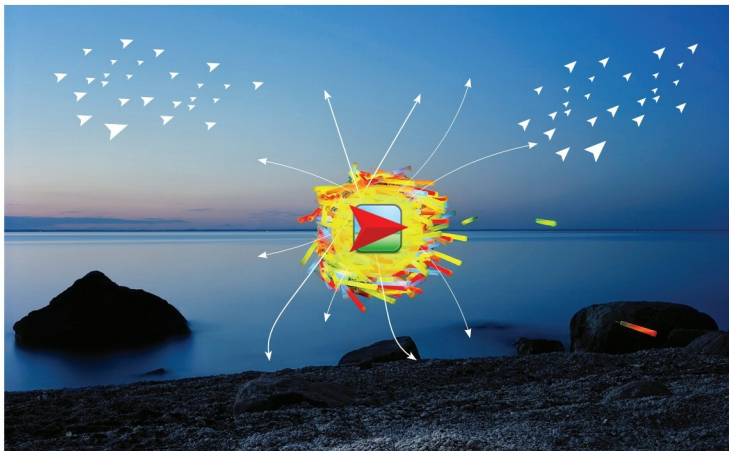


NetLogo: An Open Source Platform for Studying Emergent Behaviour

Here's a brief note on NetLogo, which can be described as a multi-agent programmable modelling environment. It is widely used by teachers, students and researchers to model and study complex systems by creating simulations and 'playing' with them.



Optimisation is inherent in nature. We have seen the key aspects of optimisation in previous editions of *OSFY*. Another interesting phenomenon is called emergent behaviour, which could be natural or artificial. Swarms of ants, schools of fish, flocks of birds, etc, are some of the natural examples of emergent behaviour. The increased complexity of routing air traffic is considered to be artificial emergent behaviour. Studying these complex behaviours using numerical techniques is not easy. Hence, an agent based simulation platform called NetLogo is used to understand such phenomena.

Introducing NetLogo

NetLogo is GNU GPL licensed free software, designed by Uri Wilensky, director of Northwestern University, in 1999. It is a JVM based cross-platform multi-agent simulation platform. NetLogo doesn't require coding knowledge or any other prerequisites. It is designed for a broad audience. So anyone

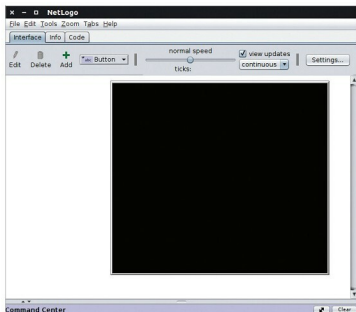


Figure 1: NetLogo interface 1